

Data Analytics Capstone

Anomaly-Based Intrusion Detection Using Machine Learning and Deep Learning

Synopsis

Ashvini. Sugur

Walsh College

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Vikas.S

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Introduction

1.1 Background and Context

The rapid proliferation of internet-connected systems has transformed the cybersecurity landscape into one of the most critical operational challenges facing modern organisations. The global cybercrime cost is projected to reach USD 10.5 trillion annually by 2025, up from USD 3 trillion in 2015, representing the largest transfer of economic wealth in history (Morgan, 2020). Every sector banking, healthcare, critical infrastructure, defence, and retail is now exposed to an increasingly sophisticated and persistent threat landscape. As digital transformation accelerates through cloud migration, remote work, and the Internet of Things (IoT), the attack surface available to adversaries has expanded dramatically. By 2025, over 75 billion IoT devices are estimated to be connected globally, each representing a potential entry point for malicious actors (Cisco, 2023).

Traditional Intrusion Detection Systems (IDS) have historically relied on signature-based detection, which matches observed network traffic against a database of known attack patterns. While effective against catalogued threats, signature-based systems are fundamentally incapable of detecting novel, zero-day, or polymorphic attacks precisely the categories of threats that are increasing most rapidly. In 2023 alone, over 25,000 new Common Vulnerabilities and Exposures (CVEs) were published, a record high, and cybercriminals routinely modify known malware to evade signature databases. Furthermore, more than 90% of internet traffic is now encrypted via HTTPS, rendering payload-inspection techniques largely ineffective (Verizon, 2023). These

limitations have created an urgent demand for adaptive, intelligent detection approaches that do not depend on prior knowledge of attack signatures.

Anomaly-based Intrusion Detection Systems (AIDS) offer a fundamentally different paradigm: they learn a statistical model of normal network behaviour and flag deviations from that model as potential intrusions. This approach enables the detection of previously unseen attacks, insider threats, and zero-day exploits categories that signature-based systems are blind to. Machine Learning (ML) and Deep Learning (DL) techniques have emerged as the dominant methods for implementing AIDS, owing to their ability to learn complex, high-dimensional patterns from large volumes of network traffic data automatically. Companies such as Darktrace, CrowdStrike, and Vectra AI have built commercially successful products on anomaly-based ML detection, validating the approach's real-world viability (Gartner, 2024).

The Canadian Institute for Cybersecurity Intrusion Detection Evaluation Dataset 2017 (CICIDS-2017), developed by the University of New Brunswick (UNB), represents the current benchmark for evaluating anomaly-based IDS research. Unlike older datasets such as KDD Cup 1999, which are now considered obsolete and unrepresentative, CICIDS-2017 was generated from realistic network traffic over five working days in July 2017, incorporating both benign traffic and 14 contemporary attack categories including DoS, DDoS, PortScan, Brute Force, Botnet, Infiltration, Web Attacks, and Heartbleed (Sharafaldin et al., 2018). The dataset presents several research challenges that reflect real-world deployment conditions: extreme class imbalance, high-dimensional feature spaces, infinite and missing values in certain

features, and temporal variation across days, all of which must be addressed rigorously in any credible study.

1.2 Problem Statement

Despite significant research activity on anomaly-based IDS, critical gaps remain in the literature that limit the practical applicability of proposed systems. Most existing studies evaluate a single model in isolation, without rigorous statistical comparison across multiple ML and DL approaches on the same standardized test set. Class imbalance particularly extreme for rare attacks such as Heartbleed, which has only 11 samples in CICIDS-2017 is frequently ignored or handled inconsistently, leading to inflated accuracy metrics that mask poor detection rates for the most dangerous attacks. Furthermore, the majority of published models operate as black boxes: they report accuracy figures without explaining which network flow features drive their decisions, making the models untrustworthy from an operational security perspective. Finally, temporal generalisation whether a model trained on Monday's traffic can detect Friday's attacks is rarely evaluated, leaving a critical question about real-world concept drift unanswered. This project addresses all four of these gaps through a rigorous, reproducible, and explainable experimental framework applied to the CICIDS-2017 dataset.

1.3 Purpose of the Study

This study aims to design, implement, evaluate, and compare a suite of anomaly-based IDS models spanning unsupervised (Isolation Forest), supervised classical (Random Forest, XGBoost), and deep learning (Long Short-Term Memory

[LSTM] Autoencoder) approaches that are applied to the CICIDS-2017 network traffic dataset. Specifically, the study seeks to:

- (a) classify network flows as either benign or one of 14 attack types;
- (b) quantify the impact of class imbalance handling techniques on rare-attack detection;
- (c) explain model decisions through SHapley Additive exPlanations (SHAP)-based feature importance analysis;
- (d) evaluate cross-day generalisation to assess the presence and magnitude of concept drift in network intrusion data.

Scope and Objectives

This study is scoped to the CICIDS-2017 dataset, which contains approximately 2.8 million labeled network flow records. The analysis is limited to the 78 numerical features extracted by CICFlowMeter and does not involve raw packet-level (PCAP) analysis. The study covers binary classification (benign vs. attack) and multi-class classification across all 14 attack categories. The following four research questions define the analytical scope:

RQ1 — Algorithm Performance Comparison

Research Question: Do deep learning models (LSTM Autoencoder) achieve significantly higher anomaly detection performance than traditional machine learning models (Random Forest, XGBoost) on the CICIDS-2017 dataset, as measured by F1-score and Precision-Recall Area Under Curve (PR-AUC)?

Null Hypothesis (H_{01}): There is no statistically significant difference in anomaly detection performance (F1-score, PR-AUC) between deep learning models and traditional machine learning models on the CICIDS-2017 dataset.

Alternate Hypothesis (H_{11}): Deep learning models achieve significantly higher anomaly detection performance (F1-score, PR-AUC) compared to traditional machine learning models on the CICIDS-2017 dataset.

Statistical Test: Wilcoxon signed-rank test applied to cross-validated F1-scores across five folds, with significance level $\alpha = 0.05$.

RQ2 — Class Imbalance Mitigation

Research Question: Does the application of Synthetic Minority Over-sampling Technique (SMOTE) significantly improve the recall and F1-score for minority attack classes (Heartbleed, Infiltration, Web Attacks) in the CICIDS-2017 dataset compared to a baseline model trained without any balancing?

Null Hypothesis (H_{02}): SMOTE does not significantly improve recall and F1-score for minority attack classes compared to a baseline model trained without class balancing.

Alternate Hypothesis (H_{12}): Applying SMOTE significantly improves recall and F1-score for minority attack classes compared to a baseline model trained without balancing.

Statistical Test: McNemar's test on per-class prediction matrices, comparing baseline vs. SMOTE-augmented model predictions on the same held-out test set.

RQ3 — Feature Importance and Explainability

Research Question: Do a small subset of network flow features contribute disproportionately to anomaly detection decisions in CICIDS-2017, and does a model trained on only the top-K SHAP-identified features perform comparably to a full-feature model?

Null Hypothesis (H_{03}): All 78 network flow features contribute equally to anomaly detection decisions, and no subset of features is disproportionately responsible for distinguishing attack from benign traffic.

Alternate Hypothesis (H_{13}): A small subset of network flow features (identified via SHAP values) contributes disproportionately to detection decisions, and a model trained on only those features performs comparably to the full-feature model.

Statistical Test: Paired t-test across five cross-validation folds comparing full-feature model F1-score vs. top-K feature model F1-score ($K = 10, 15, 20$).

RQ4 — Generalisation and Concept Drift

Research Question: Do anomaly detection models trained on one day's network traffic in CICIDS-2017 exhibit a statistically significant performance drop when evaluated on a different day's traffic, indicating the presence of concept drift?

Null Hypothesis (H_{04}): There is no significant difference in anomaly detection performance when models are evaluated on the same day's traffic (in-distribution) versus a different day's traffic (out-of-distribution).

Alternate Hypothesis (H_{14}): Models trained on one day's traffic show a statistically significant drop in performance when evaluated on a different day's traffic, confirming the presence of concept drift.

Statistical Test: Paired t-test applied across multiple train-day / test-day combinations, measuring F1-score degradation with $\alpha = 0.05$.

Sample Size Calculation

Sample size adequacy is established using the Power of Test method for each research question, employing the formula $n \geq 50 + 8m$, where m represents the number of predictor features used per model. For all four research questions, $m = 8$ key features are selected as a conservative estimate, yielding a minimum sample size of $n \geq 50 + (8 \times 8) = 114$ observations per class per experiment. For hypothesis testing via the Wilcoxon signed-rank test and McNemar's test, standard parameters of $\alpha = 0.05$ (two-tailed), $\beta = 0.20$ (i.e., Power = 0.80), and a medium effect size (Cohen's $d = 0.50$) are applied.

Research Question	Method Used	Key Parameters	Minimum N
RQ1 — Model comparison	Power of Test	$\alpha = 0.05$, Power = 0.80, $d = 0.50$, $m = 8$	$n \geq 114$ per class (Total: ~20,000)

RQ2 — Imbalance	Power of Test	$\alpha = 0.05$, Power = 0.80, McNemar correction, SMOTE applied to minority classes only	$n \geq 114$ per minority class (SMOTE augments to 500+)
RQ3 — SHAP features	Power of Test	$\alpha = 0.05$, Power = 0.80, paired t-test, $K \in$ $\{10, 15, 20\}$	$n \geq 114$ per class (SHAP on 1,000/class)
RQ4 — Concept drift	Power of Test	$\alpha = 0.05$, Power = 0.80, paired t-test across day combinations	$n \geq 114$ per class per day (500/class stratified)

Table 1. Sample size calculation summary for all four research questions.

To ensure adequate statistical power across all analyses, the final sample size is selected as the maximum of the minimum sample sizes obtained across all research questions. A stratified sampling strategy is applied: majority classes (BENIGN, DoS Hulk, PortScan) are capped at 5,000 samples each, while all minority classes are used in full. For Heartbleed (11 samples), SMOTE augmentation generates 500 synthetic samples for training, ensuring the $n \geq 114$ floor is satisfied. The final working dataset contains approximately 25,000 records across 15 classes, with an 80:20 stratified train-test split.

Data Description

4.1 Data Source

The primary dataset used in this study is the CICIDS-2017 dataset, developed by the Canadian Institute for Cybersecurity (CIC) at the University of New Brunswick (UNB), Canada. The dataset is publicly available at the following URL:

<https://www.unb.ca/cic/datasets/ids-2017.html>

The dataset was originally described in the paper: Sharafaldin, I., Lashkari, A. H., & Ghorbani, A. A. (2018). Toward generating a new intrusion detection dataset and intrusion traffic characterization. In Proceedings of the 4th International Conference on Information Systems Security and Privacy (ICISSP), Portugal.

4.2 Dataset Overview

The CICIDS-2017 dataset consists of labeled network flow records captured over five consecutive working days (Monday 3 July 2017 to Friday 7 July 2017) within a simulated enterprise network environment at UNB. Network traffic was generated using CICFlowMeter, which extracts 78 numerical statistical features from raw packet captures (PCAPs). Each record represents one bidirectional network flow — a complete conversation between two hosts labeled as either BENIGN or one of 14 attack categories.

Attribute	Value	Attribute	Value
Total records	~2,830,743	Feature count	78 numerical + 1 label

Capture period	5 days (July 2017)	Label type	Categorical (15 classes)
CSV files	8 files (one per session)	Feature extractor	CICFlowMeter
Benign samples	~2,273,097 (80.3%)	Rarest attack	Heartbleed (11 records)

Table 2. CICIDS-2017 dataset overview.

4.3 Data Dictionary — Key Features

The following table presents a representative subset of the 78 features, covering the most diagnostically important variables used in anomaly detection (see Appendix A for the complete data dictionary):

Feature Name	Type	Range	Description
Flow Duration	Numeric	0 – ∞	Total duration of the bidirectional flow in microseconds
Fwd Packet Length Mean	Numeric	0 – ∞	Mean size (bytes) of packets sent in the forward direction
Bwd Packet Length Mean	Numeric	0 – ∞	Mean size (bytes) of packets received in the backward direction

Flow Bytes/s	Numeric	0 – ∞	Total bytes transferred per second; high values signal DoS/DDoS
Flow Packets/s	Numeric	0 – ∞	Total packets per second; combined with duration flags floods
Flow IAT Mean	Numeric	0 – ∞	Mean inter-arrival time between packets; regularity signals botnet C2
RST Flag Count	Integer	0 – n	Count of TCP RST flags; high count signals PortScan activity
SYN Flag Count	Integer	0 – n	Count of TCP SYN flags; excessive SYN indicates SYN flood
Init Win Bytes Fwd	Integer	0 – 65535	Initial TCP window size; anomalous values flag Slowloris
Fwd Packets/s	Numeric	0 – ∞	Forward direction packet rate; primary Brute Force signal
Bwd Packet Length Max	Numeric	0 – ∞	Maximum backward packet size; large values signal data exfiltration
Label	Categorical	15 classes	Ground truth: BENIGN or named attack type

Table 3. Data dictionary — representative feature subset from CICIDS-2017 (see Appendix A for complete list).

4.4 Class Distribution

The dataset exhibits severe class imbalance, characteristic of real-world network environments where attack traffic represents a small minority of total flows. Table 4 presents the class distribution across the full dataset:

Attack Category	Sample Count	% of Dataset	Day Captured
BENIGN	2,273,097	80.30%	Mon–Fri
DoS Hulk	231,073	8.16%	Wednesday
PortScan	158,930	5.62%	Friday
DDoS	128,027	4.52%	Friday
DoS GoldenEye	10,293	0.36%	Wednesday
FTP-Patator	7,938	0.28%	Tuesday
SSH-Patator	5,897	0.21%	Tuesday
DoS Slowloris	5,796	0.20%	Wednesday
DoS Slowhttptest	5,499	0.19%	Wednesday
Bot	1,966	0.07%	Friday
Web Attack – Brute Force	1,507	0.05%	Thursday

Web Attack – XSS	652	0.02%	Thursday
Infiltration	36	0.001%	Thursday
Web Attack – SQL Injection	21	0.001%	Thursday
Heartbleed	11	0.0004%	Wednesday

Table 4. Class distribution across the full CICIDS-2017 dataset.

Analytic Approach

5.1 Data Preprocessing Pipeline

All eight CSV files are merged into a single dataframe. Column names are stripped of leading and trailing whitespace. Infinite values in Flow Bytes/s and Flow Packets/s (caused by division by near-zero flow durations) are replaced with NaN and subsequently removed. Duplicate rows are eliminated. Features are normalised to the [0, 1] range using Min-Max scaling. Binary and multiclass label encodings are created. An 80:20 stratified train-test split is applied, preserving class proportions in both partitions.

5.2 Solution to RQ1 — Model Comparison

RQ1 is addressed through a standardised comparative experiment. Five models are trained on the same training set and evaluated on the same held-out test set using five-fold stratified cross-validation: (1) Isolation Forest (unsupervised baseline), (2)

Random Forest (supervised classical ML), (3) XGBoost (supervised gradient boosting), and (4) LSTM Autoencoder (deep learning). The primary evaluation metric is the macro-averaged F1-score; secondary metrics include PR-AUC and per-class recall. The Wilcoxon signed-rank test is applied to the five cross-validated F1-scores of each model pair to determine whether performance differences are statistically significant ($\alpha = 0.05$).

The F1-score, PR-AUC, Precision, and Recall are computed as follows:

$$\text{Precision} = TP / (TP + FP); \text{ Recall} = TP / (TP + FN); F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

where TP = True Positives, FP = False Positives, FN = False Negatives. For multiclass classification, macro-averaging computes each metric independently per class and averages without weighting by class size, ensuring rare attack categories are not dominated by the majority class.

5.3 Solution to RQ2 — Class Imbalance

RQ2 is addressed by training Random Forest under two conditions on the same features and train-test split: (a) Baseline: no balancing applied, and (b) SMOTE-augmented: SMOTE is applied to the training set only, generating synthetic minority samples using $k = 5$ nearest neighbours. The test set remains unmodified in both conditions. Per-class F1-score and recall are recorded for Heartbleed, Infiltration, Web Attack (SQL Injection), and Web Attack (XSS). McNemar's test is applied to the binary prediction vectors (correct/incorrect) for each minority class to determine whether the improvement from SMOTE is statistically significant.

McNemar's test statistic is computed as:

$$\chi^2 = (b - c)^2 / (b + c)$$

where b = number of flows correctly classified by SMOTE model but not baseline, c = number correctly classified by baseline but not SMOTE model. A p -value < 0.05 leads to rejection of H_{02} .

5.4 Solution to RQ3 — SHAP Feature Importance

RQ3 is addressed in two stages. First, SHAP values are computed for the best-performing model identified in RQ1 using a stratified sample of 1,000 flows per class. The mean absolute SHAP value for each of the 78 features is computed, producing a ranked list of feature importance. Second, the model is retrained using only the top 10, top 15, and top 20 features identified by SHAP, and the resulting F1-scores are compared against the full-78-feature model using a paired t-test across five cross-validation folds. A non-significant performance drop with the reduced feature set supports H_{13} .

The SHAP value for feature i in sample x is defined as:

$$\varphi_i = \sum [|S|!(p-|S|-1)!/p!] \times [f(S \cup \{i\}) - f(S)]$$

where the sum is over all subsets S of features excluding feature i , p is the total number of features, and $f(S)$ is the model output using only features in S .

5.5 Solution to RQ4 — Concept Drift

RQ4 is addressed using a cross-day temporal evaluation design. The best-performing supervised model from RQ1 is trained on Monday’s data (benign only, as a day without attacks) augmented with attack samples from Tuesday (FTP-Patator, SSH-Patator). The model is then evaluated on: (a) Monday held-out samples (in-distribution), and (b) Friday samples (DDoS, PortScan, Bot — out-of-distribution). This process is repeated for all five-day combinations, producing a distribution of in-distribution vs. out-of-distribution F1-scores. A paired t-test on these distributions tests H_{14} : that a statistically significant performance drop exists when evaluating out-of-distribution.

5.6 Evaluation Metrics Summary

RQ	Primary Metric	Statistical Test	Threshold
RQ1	Macro F1, PR-AUC	Wilcoxon signed-rank test	$\alpha = 0.05$
RQ2	Per-class F1, Recall	McNemar’s test	$\alpha = 0.05$
RQ3	F1 (full vs. top-K)	Paired t-test (5-fold)	$\alpha = 0.05$
RQ4	F1 drop (in vs. out)	Paired t-test (days)	$\alpha = 0.05$

Table 5. Evaluation metrics and statistical tests for each research question.

Recommendations and Applications

6.1 Target Users

The primary users of the findings and prototype system developed in this project are network security operations centre (SOC) analysts, cybersecurity engineers, and IT security managers in medium-to-large enterprises across banking, healthcare, critical infrastructure, and e-commerce sectors. Academic researchers working on intrusion detection, network security, and applied machine learning represent a secondary user group. Policymakers and compliance officers working under frameworks such as GDPR, HIPAA, NIS2, and India's CERT-In regulations constitute a tertiary audience, as the project's findings directly inform the selection and justification of automated intrusion detection tools for regulatory compliance.

6.2 Practical Recommendations

Based on the experimental design and anticipated findings, the following recommendations are proposed for practitioners:

- Deploy ensemble or hybrid models (combining Random Forest with an Autoencoder) for production IDS, as they offer both high accuracy on known attacks and sensitivity to novel anomalies without requiring attack labels.
- Always apply SMOTE or class-weighted training when deploying IDS on real-world networks, where rare attack types are precisely the most dangerous and most likely to be missed by unbalanced models.

- Integrate SHAP-based explainability into SOC dashboards, enabling analysts to see not just that a flow is flagged as malicious, but which specific features (e.g., RST Flag Count, Flow Bytes/s) triggered the alert.
- Retrain IDS models periodically to account for concept drift, as network traffic patterns evolve with new applications, protocols, and attack techniques.
Cross-day evaluation results provide an empirical basis for recommending retraining frequency.
- Use a lightweight Streamlit or Flask dashboard prototype as the operational interface, enabling real-time scoring of incoming network flows with per-flow anomaly explanations.

References

- Alqahtani, A., & AlShaher, H. (2024). Anomaly-based intrusion detection systems using machine learning. *Journal of Cybersecurity and Information Management*, *14*(1), 20–33. <https://doi.org/10.54216/JCIM.140102>
- Al-madani, A. M. (2024). Enhanced intrusion detection with LSTM-based model, feature selection, and SMOTE for imbalanced data. *Applied Sciences*, *14*(2), 479. <https://doi.org/10.3390/app14020479>
- Buczak, A. L., & Guven, E. (2016). A survey of data mining and machine learning methods for cyber security intrusion detection. *IEEE Communications Surveys & Tutorials*, *18*(2), 1153–1176. <https://doi.org/10.1109/COMST.2015.2494502>
- Cisco Systems. (2023). Cisco annual internet report (2018–2023). Cisco. <https://www.cisco.com/c/en/us/solutions/collateral/executive-perspectives/annual-internet-report/white-paper-c11-741490.html>
- Gartner. (2024). Market guide for network detection and response. Gartner Research.
- Liu, H., & Lang, B. (2019). Machine learning and deep learning methods for intrusion detection systems: A survey. *Applied Sciences*, *9*(20), 4396. <https://doi.org/10.3390/app9204396>
- Maseer, Z. K., Yusof, R., Bahaman, N., Mostafa, S. A., & Foozy, C. F. M. (2021). Benchmarking of machine learning for anomaly based intrusion detection systems in the CICIDS2017 dataset. *IEEE Access*, *9*, 22351–22370. <https://doi.org/10.1109/ACCESS.2021.3056614>

Morgan, S. (2020). Cybercrime to cost the world \$10.5 trillion annually by 2025. Cybercrime Magazine.

<https://cybersecurityventures.com/hackerpocalypse-cybercrime-report-2016/>

Rahman, M. M. U. (2023). Multi-class network intrusion detection with class imbalance via LSTM and SMOTE. arXiv preprint arXiv:2310.01850. <https://arxiv.org/abs/2310.01850>

Sharafaldin, I., Lashkari, A. H., & Ghorbani, A. A. (2018). Toward generating a new intrusion detection dataset and intrusion traffic characterization. In Proceedings of the 4th International Conference on Information Systems Security and Privacy (ICISSP) (pp. 108–116). SCITEPRESS. <https://doi.org/10.5220/0006639801080116>

Singh, P., Nehra, D., Mangat, V., & Kumar, K. (2025). Enhancing intrusion detection with ResNet and SMOTE-ENN: A deep learning approach to class imbalance in CICIDS2017. International Journal of Information Technology. <https://doi.org/10.1007/s41870-025-02766-9>

Verizon. (2023). 2023 data breach investigations report. Verizon Business.

<https://www.verizon.com/business/resources/reports/dbir/>

Zhao, G.,(2024). Explainable deep learning-based feature selection and intrusion detection method on the Internet of Things. Sensors, 24(16), 5223. <https://doi.org/10.3390/s24165223>

Zhong, M.,(2025). Improved network anomaly detection system using optimized autoencoder-LSTM. Expert Systems with Applications.

<https://doi.org/10.1016/j.eswa.2025.127462>

Frontiers in Artificial Intelligence. (2025). A deep learning/machine learning approach for anomaly based network intrusion detection. Frontiers in Artificial Intelligence.

<https://doi.org/10.3389/frai.2025.1625891>

Appendix A: Complete CICIDS-2017 Feature List

The following table lists all 78 features extracted by CICFlowMeter from the CICIDS-2017 dataset. This appendix is referenced in Section 4.3 (Data Dictionary). Features marked with an asterisk (*) are among the top features identified in prior literature for anomaly detection (Maseer et al., 2021).

#	Feature Name	#	Feature Name	#	Feature Name
1	Flow Duration*	27	Bwd Packet Length Std	53	Idle Mean
2	Total Fwd Packets	28	Flow Bytes/s*	54	Idle Std
3	Total Bwd Packets	29	Flow Packets/s*	55	Idle Max
4	Total Length Fwd Packets	30	Flow IAT Mean*	56	Idle Min
5	Total Length Bwd Packets	31	Flow IAT Std	57	Fwd Avg Bytes/Bulk
6	Fwd Packet Length Max	32	Flow IAT Max	58	Fwd Avg Packets/Bulk

7	Fwd Packet Length Min	33	Flow IAT Min	59	Fwd Avg Bulk Rate
8	Fwd Packet Length Mean*	34	Fwd IAT Total	60	Bwd Avg Bytes/Bulk
9	Fwd Packet Length Std	35	Fwd IAT Mean	61	Bwd Avg Packets/Bulk
10	Bwd Packet Length Max*	36	Fwd IAT Std	62	Bwd Avg Bulk Rate
11	Bwd Packet Length Min	37	Fwd IAT Max	63	Subflow Fwd Packets
12	Bwd Packet Length Mean*	38	Fwd IAT Min	64	Subflow Fwd Bytes
13	Bwd Packet Length Std	39	Bwd IAT Total	65	Subflow Bwd Packets
14	Flow Bytes/s*	40	Bwd IAT Mean	66	Subflow Bwd Bytes
15	Flow Packets/s*	41	Bwd IAT Std	67	Init Win Bytes Fwd*
16	Flow IAT Mean*	42	Bwd IAT Max	68	Init Win Bytes Bwd
17	Flow IAT Std	43	Bwd IAT Min	69	Act Data Pkt Fwd

18	Flow IAT Max	44	Fwd PSH Flags	70	Min Seg Size Fwd
19	Flow IAT Min	45	Bwd PSH Flags	71	Active Mean
20	Fwd Packets/s*	46	Fwd URG Flags	72	Active Std
21	Bwd Packets/s	47	Bwd URG Flags	73	Active Max
22	Packet Length Min	48	Fwd Header Length	74	Active Min
23	Packet Length Max	49	Bwd Header Length	75	Idle Mean
24	Packet Length Mean	50	FIN Flag Count	76	Idle Std
25	Packet Length Std	51	SYN Flag Count*	77	Idle Max
26	Packet Length Variance	52	RST Flag Count*	78	Label (target)

Table A1. Complete list of 78 features in CICIDS-2017. Features marked () are top-ranked in prior anomaly detection literature.*